

***Effect of transport in heterogenous catalytic reaction systems

*Heterogenous catalytic reactions involves diffusion of species from bulk to the surface of catalytic where they react.

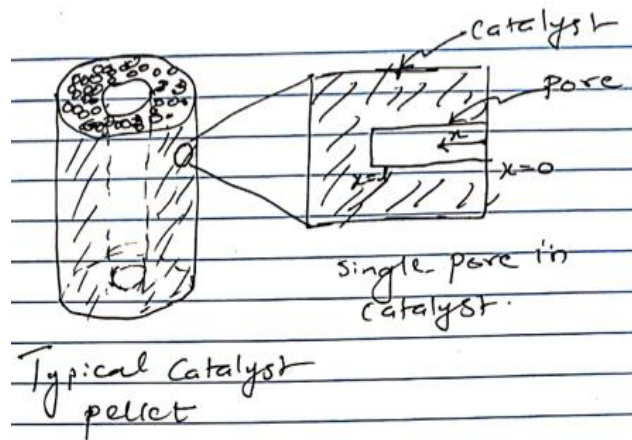
*Catalyst are usually compacted into pellets consisting of several connected pores yielding high surface area.

*Reactions must be transported (via diffusion) from the bulk to the surface of pores

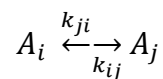
*The accessibility of reactants to the pores govern the overall rate of reaction.

*Our objective is to quantify the effect of diffusion on the rate of catalytic reaction.

*Consider a cylindrical pore of length 'l' and cross-sectional area 'A'.



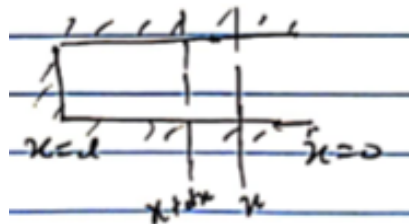
*Consider multiple first-order reactions occurring on catalyst surface



*Rate of reaction is defined as

$$\hat{r}_{ji} = \hat{k}_{ji} C_i \quad (140)$$

*Write mole balance in a shell of thickness dx



Accumulation = in – out + generation

$$\frac{dC_j}{dt} = \left[AD_j \frac{dC_j}{dt} \right]_x - \left[AD_j \frac{dC_j}{dt} \right]_{x+dx} + Pdx \sum_{i=1}^n \alpha_{ij} \hat{r}_{ij}, \quad j = 1, 2, \dots, n$$

P = perimeter of pore

Assuming no accumulation and taking limit

$$\underline{D} \frac{d^2 \underline{C}}{dx^2} + \frac{P}{A} \underline{\hat{K}} \underline{C} = 0 \quad (141)$$

Boundary conditions

At $x=0$, $\underline{C} = \underline{C}_0$ [fixed bulk the concentration]

At $x=l$, $d\underline{C}/dx = 0$ [no flux at the end of pore]

De-dimensionalize the equations using dimensionless length

$$\xi = \frac{x}{l} \quad (143)$$

And dimensionless matrix Thiele Modules

$$\underline{\Phi}^2 = -\frac{Pl^2}{A} \underline{D}^{-1} \underline{\hat{K}} \quad (144)$$

Scalar Thiele modules can be represented as

$$\Phi^2 = \frac{Pl^2}{A} \frac{K}{D} = \frac{\text{Reaction Rate}}{\text{Diffusion Rate}} \quad (145)$$

We can rewrite eq. (141) as

$$\frac{d^2 \underline{C}}{d\xi^2} = \underline{\Phi}^2 \underline{C} \quad (146)$$

At $\xi=0$, $\underline{C} = \underline{C}_0$, at $\xi=1$, $d\underline{C}/d\xi=0$ (147)

Note $\underline{\Phi}^2$ is a positive definite and singular matrix

➔ Taking inspiration from the solution of scalar eq. of type (146), we can write solution for vector eq. as

$$\underline{C} = [\cosh \Phi(1 - \xi)] \underline{u}_1 + [\sinh \Phi(1 - \xi)] \underline{u}_2 \quad (148)$$

Apply boundary condition at $\xi=1 \rightarrow \underline{u}_2 = 0$

$$\underline{C} = [\cosh\Phi(1 - \xi)]\underline{u}_1$$

Apply boundary condition at $\xi=0$

$$\underline{u}_1 = [\cosh\Phi]^{-1}C_o$$

Solution can be written as

$$\underline{C} = [\cosh\Phi(1 - \xi)][\cosh\Phi]^{-1}C_o \quad (149)$$

(note that Φ is singular but $\cosh\Phi$ is not)

The actual reaction rate is given by the flux of species entering at the mouth of pore.

$$\text{Actual reaction rate} = -A \underline{D} \frac{dC}{dx} \Big|_{x=0} = -\frac{A}{l} \underline{D} \frac{dC}{d\xi} \Big|_{\xi=0} \quad (150)$$

In absence of mass-transport effects, the entire pore surface will have uniform concentration C_o . The rate of reaction in absence of mass-transfer is given as $-Pl\hat{K}C_o$ (151)

Now we can define effectiveness factor = (reaction rate in presence of mass transfer)/(Reaction rate in absence of mass transfer)

$$\rightarrow \underline{Npl}\hat{K}C_o = \frac{A}{l} \underline{D} \frac{dC}{d\xi} \Big|_{\xi=0} = -\frac{A}{l} \underline{D} \Phi [\sinh\Phi] [\cosh\Phi]^{-1} C_o = -\frac{A}{l} \underline{D} \Phi [\tanh\Phi] C_o \quad (152)$$

$$\rightarrow \underline{N}\hat{K} = \frac{A}{pl^2} \underline{D} \Phi [\tanh\Phi] = \hat{K} \Phi^{-1} [\tanh\Phi] \quad (153)$$

For single reaction, $N = \frac{\tanh\Phi}{\Phi}$ (154). If we define the rate of reaction in absence of mass transfer as $\hat{K}N'C_o$, then the effectiveness factor can be expressed as

$$\underline{N}' = \Phi^{-1} [\tanh\Phi] \quad (155)$$

This is similar to eqn. (154)

Recall: solution of $\frac{dx}{dt} = \underline{kx} \cdot \underline{x} = e^{kt} \underline{x}_0 = \underline{x}^* + \sum_{j=1}^{n-1} e^{\lambda_j t} \frac{\underline{z}_j^T \underline{D}^{-1} \underline{x}_0}{\underline{z}_j^T \underline{D}^{-1} \underline{z}_j} \underline{z}_j$

$$\underline{x} = \underline{x}^* + \sum_{j=1}^{n-1} e^{\lambda_j t} \frac{(\underline{D}^{-1} \underline{z}_j)^T}{\underline{z}_j^T \underline{D}^{-1} \underline{z}_j} \underline{x}_0 = \sum_{j=1}^{n-1} e^{\lambda_j t} \frac{\underline{z}_j (\underline{D}^{-1} \underline{z}_j)^T}{\underline{z}_j^T (\underline{D}^{-1} \underline{z}_j)} \underline{x}_0$$

$$\frac{\underline{z}_j (\underline{D}^{-1} \underline{z}_j)^T}{\underline{z}_j^T (\underline{D}^{-1} \underline{z}_j)} = \underline{P}_j$$

$\underline{P}_j$ is the projection matrix corresponding to a matrix $\underline{t}$.

$\underline{z}_j$ is the column eigen vector of $\underline{K}$

$\underline{D}^{-1} \underline{z}_j$ is the row eigen vector of $\underline{K}$

$$e^{\underline{k}t} = \sum_{j=0}^{n-1} e^{\lambda_j t} \underline{P}_j$$

In general, any function of matrix can be written as

$$f(\underline{k}) = \sum_{j=0}^{n-1} \delta(\lambda_j) \underline{P}_j \quad (156)$$

The effectiveness factor can be written as

$$\underline{N}' = \Phi^{-1}[\tanh\Phi] = \underline{P}_0 + \sum_{j=1}^{n-1} \lambda_j^{-1} \tanh\lambda_j \underline{P}_j \quad (157)$$

Lets consider two extreme limits of Φ

1. $\Phi \rightarrow 0$ Thiele modules of all reactions are very small.
All the eigenvalues of Φ are very small, $\lambda_j \rightarrow 0$

$$\lim_{\lambda_j \rightarrow 0} \frac{\tanh \lambda_j}{\lambda_j} = 1$$

Substitute in (157)

$$\underline{N}' = \sum_{j=0}^{n-1} \underline{P}_j = \underline{I}$$

→ effectiveness factor of all reactions are unity. [no transport limitations]

2. $\Phi \gg 0$ Thiele modules of all reactions are very large.
All eigenvalues are very large.

$$\lim_{\lambda_j \gg 0} \frac{\tanh \lambda_j}{\lambda_j} \sim 0$$

$$\underline{N}' = \underline{P}_0$$

Effectiveness factors are less than unity. All reactions are close to equilibrium.

***Effect of Temperature on the Rate of reaction and Effectiveness factor

*Consider a single first order reaction

*The effectiveness factor is given by (154)

$$\begin{aligned} \underline{N} &= \frac{\tanh\Phi}{\Phi} = \frac{\text{reaction rate with mass transfer}}{\text{reaction rate without mass transfer}} \\ &= \frac{\hat{r}_{obs} \text{ (measured or observed)}}{\hat{r} \text{ (intrinsic or actual)}} \\ \hat{r}_{obs} &= N \hat{r} \quad (157) \end{aligned}$$

The thiele modules for a single reaction in a single pore is

$$\Phi^2 = \frac{PlKC}{\frac{ADC}{L}}$$

Thiele modules for a single reaction occurring over catalyst pellet (containing multiple pores) can be defined as

$$\Phi^2 = \frac{\rho_g S_g \hat{k}(T) a^2}{De} \quad (158)$$

ρ_g =bulk density of pellet

a = characteristic length of pellet

De = Effective diffusion coefficient of reactant in pellet

S_g = surface area of catalyst per unit mass

For a first order reaction

$$\hat{r} = \hat{k}(T)C = Ae^{-\frac{E}{RT}}C$$
$$\ln \hat{r} = \ln AC - \frac{E}{RT}$$

Take derivative w.r.t T^{-1}

$$\frac{d \ln \hat{r}}{dT^{-1}} = \frac{d \ln \hat{k}}{dT^{-1}} = \frac{-E}{R} \quad (159)$$

This is the variation of intrinsic rate w.r.t temperature.

From (157)

$$\hat{r}_{obs} = N\hat{r}$$
$$\ln \hat{r}_{obs} = \ln \hat{r} + \ln N$$
$$\frac{d \ln \hat{r}_{obs}}{dT^{-1}} = \frac{d \ln \hat{r}}{dT^{-1}} + \frac{d \ln N}{dT^{-1}} = \frac{-E}{R} + \frac{d \ln N}{d \ln \Phi} \frac{d \ln \Phi}{dT^{-1}} \quad (160)$$

For large enough Φ

$$N = \frac{1}{\Phi}$$
$$\frac{dN}{d\Phi} = \frac{-1}{\Phi^2}$$
$$\frac{d \ln N}{d \ln \Phi} = \frac{-1}{N\Phi} = -1 \quad (161)$$

Substitute (161) to (160)

$$\frac{d \ln \hat{r}_{obs}}{dT^{-1}} = \frac{-E}{R} - \frac{d \ln \Phi}{dT^{-1}} = -\frac{[E + R d \ln \Phi / dT^{-1}]}{R}$$
$$E_{obs} = E + \frac{R d \ln \Phi}{dT^{-1}} \quad (162)$$

Substitute (158) to (162)

$$E_{obs} = E + \frac{R d \ln \hat{k}(T)}{2 dT^{-1}} = E + \frac{R}{2} \left(\frac{-E}{R} \right) = \frac{E}{2}$$

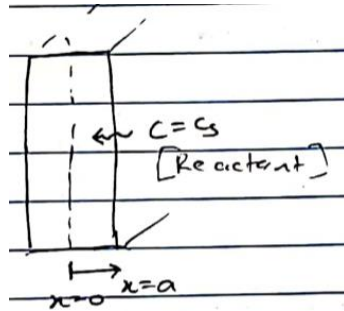
If we assume De also follows Arrhenius law, then

$$E_{obs} = E + \frac{R d \ln \hat{k}(T)}{2 dT^{-1}} - \frac{R d \ln De}{2 dT^{-1}} = E - \frac{E}{2} + \frac{E_d}{2} = \frac{E + E_d}{2}$$

The effect of mass transfer is to reduce the activation energy.

***Effectiveness factor for reactions with nonlinear kinetics

*Consider catalyst pellet with rectangular plate geometry



Also consider non-linear catalytic reaction whose rate is given as $\hat{r}(C)$

Recall mole balance for species reacting in a dispersed catalytic surface.

$$\frac{DC}{Dt} + \nabla \cdot \underline{J} = -a_v \hat{r}(C) - C \frac{D \ln \hat{V}}{Dt}$$

Consider steady state, 1D diffusion

$$\frac{d}{dx} \left[-De \frac{dC}{dx} \right] = -a_v \hat{r}(C)$$

$$De \frac{d^2 C}{dx^2} = \rho_b S_g \hat{r}(C) \quad (164)$$

B.Cs

$C=C_s$ at $x=a$

$dC/dx = 0$ at $x=0$ (165)

De-dimensionalize using : $u=C/C_s$; $\rho = x/a$, $\Phi^2 = \frac{\hat{r}(C)\rho_b S_g a^2}{De C_s}$; $R(u) = \hat{r}(C)/\hat{r}(C_s)$

Apply to eq. 164

$$\frac{d^2 u}{d\rho^2} = \Phi^2 R(u) \quad (166)$$

And B.Cs

$u=1$ at $\rho = 1$

$du/d\rho = 0$ at $\rho = 0$ (167)

From the definition of effectiveness factor

$\hat{r}(C_s)\eta = \text{reaction rate in presence of mass transfer}$

$$= \text{flux of species at the surface of pellet} = De \frac{dC}{dx} \Big|_{x=a} = De \frac{C_s}{a} \frac{du}{d\rho} \Big|_{x=a}$$

When writing η in terms of reaction rate per unit volume of catalyst

$$\rho_b S_g \hat{r}(C_s) \eta = \frac{1}{a} De \frac{C_s}{a} \frac{du}{d\rho} \Big|_{\rho=1}$$

$$\eta = \frac{1}{\Phi^2} \frac{du}{d\rho} \Big|_{\rho=1} \quad (168)$$

Let $\frac{du}{d\rho} = v$ (169)

Substitute in (166) yield

$$\frac{dv}{d\rho} = \Phi^2 R(u) \quad (170)$$

Dividing (169) by (170)

$$\frac{du}{dv} = \frac{v}{\Phi^2 R(u)}$$

$$\Phi^2 R(u) du = v dv$$

$$\int_{u_0}^u R(u) du = \frac{1}{2} \frac{v^2}{\Phi^2} \quad (171)$$

Define $F(u) = 2 \int_0^u R(x) dx$

Re-arranging (171)

$$v(\rho) = \frac{du}{d\rho} = \Phi \sqrt{F(u) - F(u_0)} \quad (172)$$

$$\int_{u_0}^1 \frac{du}{\sqrt{F(u) - F(u_0)}} = \int_0^1 \Phi d\rho = \Phi \quad (173)$$

Solve (173) for a given Φ to calculate u_0

Now effectiveness factor can be obtained as

$$\eta = \frac{1}{\Phi^2} \frac{du}{d\rho} \Big|_{\rho=1} = \frac{v(1)}{\Phi^2} = \frac{\sqrt{F(1) - F(u_0)}}{\Phi} \quad (174)$$